CyberShield – Cyber Security Awareness Platform

1. Introduction

CyberShield is a web-based cybersecurity awareness platform designed to help users identify potentially suspicious messages and URL warning signs. It also provides cybersecurity tips and an awareness quiz.

2. Problem Statement

People frequently receive suspicious messages involving fake prizes, urgent account warnings, OTP requests, and payment demands. CyberShield aims to help users recognise these warning signs.

3. Objectives

- Identify predefined suspicious patterns in messages.
- Check URLs for potentially suspicious characteristics.
- Provide cybersecurity awareness tips.
- Improve cybersecurity awareness through a quiz.

4. Technologies Used

- HTML
- CSS
- JavaScript
- Visual Studio Code
- GitHub

5. Features

Scam Message Detector

Checks user-provided messages against predefined suspicious patterns and displays a rule-based risk score.

Suspicious URL Checker

Examines selected URL characteristics and displays possible warning signs.

Cyber Safety Tips

Provides basic guidance for safer online behaviour.

Cyber Safety Quiz

Tests users' understanding of cybersecurity through a short quiz.

6. How to Run the Project

1. Open or download this GitHub repository.
2. Locate the index.html file.
3. Download the file if necessary.
4. Open index.html in a web browser.
5. Test the message detector, URL checker, and quiz.

7. Current Implementation

The current prototype uses predefined rules and JavaScript logic. It does not currently use a trained machine learning model or a real-time threat intelligence database.

8. Limitations

- Some legitimate messages may be flagged as suspicious.
- New scam patterns may not be detected.
- URL checks do not confirm whether a website is malicious.
- Risk scores are indicators, not verified probabilities.

9. Future Enhancements

- Add more detection rules.
- Improve input validation and error handling.
- Explore integration with a trusted threat intelligence API.
- Explore machine learning-based message classification.

10. Team Members

- Member 1: [Anjum S]
- Member 2: [Jyotika Priya]
- Member 3: [Dhyuti Mohan]

11. Project Status

Prototype development and testing are in progress.